MTH212M End Semester Examination
Part B (February 25, 2026)

5:00 pm - 7:00 pm (+40 minutes for DAP students) **Maximum marks for Part B: 20**

Instructions:

1. Write your answers in the designated place. If your answer is not legible, you shall not get credit.
2. DO NOT attach any additional sheet to this page.
3. Read the question stem and answer the sub-questions separately in the space provided.
4. Provide all details/justification including calculations.

Name:

Roll No.:

Question 1. Let $\{X_n\}_{n=0}^{\infty}$ be a Markov Chain with a countably infinite state space S and with the transition probability matrix $P = (p_{ij})_{i,j\in S}$. Assume that $\lim_{n\to\infty} p_{ij}^n = v_j$ for all $i, j \in S$, where v_j depends on j and not on i. Further assume that the row vector $v = (v_j)_{j\in S}$ satisfies $vP = v$.

Question 1 (a): (2 marks) Show that $\sum_{j\in S} v_j \le 1$.

Question 1 (b): (2 marks) Show that $vP^n = v$ for all $n = 2, 3, \dots$.

Question 1 (c): (2 marks) Show that either $v_j = 0$ for all $j \in S$ or $\sum_{j\in S} v_j = 1$.

Question 1 (d): (2 + 2 marks) If $v_j = 0$ for all $j \in S$, show that there is no stationary distribution. If $\sum_{j\in S} v_j = 1$, show that $v = (v_j)_{j\in S}$ is the unique stationary distribution.

Question 2. Fix $q \in (0, \frac{1}{2})$ and set $p := 1 - q$. Consider a Markov Chain with the state space $\{0, 1, \dots\}$ and transition probability matrix

$$P = \begin{pmatrix} p & q & 0 & 0 & 0 & \cdots \\ p & 0 & q & 0 & 0 & \cdots \\ 0 & p & 0 & q & 0 & \cdots \\ 0 & 0 & p & 0 & q & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Question 2 (a): (3 marks) Show that the Markov Chain is irreducible.

Question 2 (b): (2 marks) Show that the Markov Chain is aperiodic.

Question 2 (c): (5 marks) Is the Markov Chain positive recurrent?